

Executive Summary

This project proposes a scalable digital solution to address the pressing issue of postpartum depression (PPD) among new mothers in London. Social isolation, lack of emotional support, and barriers to timely healthcare access contribute significantly to maternal mental health challenges, with PPD affecting over 10–15% of new mothers annually. Current healthcare provisions are insufficient, particularly for marginalized and low-income communities, leaving a substantial care gap.

In response, we propose the development of an innovative, holistic mobile app that functions both as a peer support network and an integrated mental health intervention tool. The platform will include AI-powered chatbots for emotional check-ins, evidence-based Cognitive Behavioral Therapy (CBT) modules, virtual consultations, and community-building activities such as exercise and cooking classes. Designed with cultural sensitivity and digital accessibility in mind, the app aims to serve women who are often overlooked by traditional health services.

Technically, the solutions explore secure cloud infrastructure and OpenAI-based language models, while aligning with NHS Digital Technology Assessment Criteria (DTAC) for future integration into the UK public health system. The app is also backed by partnerships with local councils, NHS Trusts, and maternal health NGOs, ensuring clinical validity and user-centered design.

Financially, the business is supported by over £650,000 in seed and Series A funding, with a hybrid revenue model combining institutional licensing, NHS procurement, and a consumer freemium model. Projections show a compound annual revenue growth rate exceeding 150%, with breakeven anticipated by Year 3.

While challenges exist, including NHS regulatory hurdles, user retention concerns, and cultural stigma, these are mitigated through phased implementation, user-centered design, and diversified funding strategies. With the UK's femtech sector rapidly expanding, this app is well-positioned to become a first-mover in clinically integrated maternal mental health tech. Ultimately, the project seeks to enhance maternal well-being, reduce NHS burden, and contribute meaningfully to public health outcomes.

Introduction

Maternal mental health remains a critically underserved area of healthcare, with social isolation and psychological challenges, especially during the postpartum period, affecting a significant number of new mothers (NHS, 2022). In cities like London, where over 35% of the population reports frequent loneliness, many women begin their journey into motherhood without nearby family or community support (Evening Standard, 2024). For new mothers, already facing hormonal shifts, sleep deprivation, and lifestyle changes, this isolation can be overwhelming and lead to serious mental health outcomes, including postpartum/postnatal depression (PPD/PND) (Corrigan et al., 2015).

While temporary emotional disturbances or “baby blues” are common, PPD is more persistent and severe, impacting at least 1 in 10 women within the first year of childbirth (NHS, 2022). Women with a history of depression, bipolar disorder, or previous perinatal mental health issues are especially at risk (National Institute of Mental Health, 2024). PPD arises from a complex interplay of biological, psychological, and environmental factors, including the physical demands of childbirth, caregiving stress, chronic sleep deprivation and significant hormonal fluctuations, particularly the sharp drop in estrogen and progesterone after delivery, which can trigger mood instability (Trifu, Vladuti and Popescu, 2019). Alternatively, strong social support serves as one of the most effective protective factors. Mothers embedded in reliable family or community networks are less likely to develop depression, while isolation, especially when paired with economic hardship or cultural stigma, can intensify feelings of inadequacy and emotional exhaustion (Corrigan et al., 2015).

The effects of maternal PPD extend beyond the individual, influencing child development and family dynamics. Mothers often experience reduced self-esteem, emotional withdrawal, and diminished quality of life (Slomian et al., 2019). These symptoms can impair mother-infant bonding and caregiving. For instance, research has shown that children of mothers with depressive symptoms at 2–4 months postpartum are less likely to receive age-appropriate vaccinations and may face long-term developmental delays (Minkovitz et al., 2005). These findings highlight the need for early detection and timely intervention to prevent cascading negative outcomes.

From a systemic perspective, poor maternal mental health places a significant financial and logistical burden on the UK healthcare system. Mental illness more broadly is estimated to cost £118 billion annually. Within this, perinatal mental health issues cost the NHS and social

services around £1.2 billion annually (UK Government, 2019). Yet, as the British Medical Association (2024) reports, the mental healthcare system remains overstretched, facing insufficient funding, staff shortages, and limited capacity to address the social determinants of health, including poverty and domestic instability. These limitations are particularly detrimental to maternal care, where delayed diagnosis and intervention can result in long-term harm.

While screening tools like the Edinburgh Postnatal Depression Scale (EPDS) and the Postpartum Depression Screening Scale (PDSS) are effective in identifying at-risk mothers (Kucab et al., 2022), they are not universally or consistently used. Moreover, the overlap between typical postpartum symptoms and clinical depression can obscure diagnosis. Stigma also plays a role. Cultural norms that idealize maternal selflessness often prevent women from acknowledging emotional difficulties or seeking help (Modak et al., 2023; Garapati et al., 2023). The COVID-19 pandemic further exacerbated these issues by limiting face-to-face services and increasing isolation, contributing to higher rates of perinatal distress (Chen et al., 2022).

In the absence of formal care pathways, many mothers turn to social media platforms for support. While these communities can offer a sense of shared experience, they are often unmoderated, unverified, and risk amplifying distress through misinformation or judgmental interactions (Bjelke et al., 2016). On the other hand, some mental health apps have shown promise (Chan & Chen, 2019). A systematic review by Miura et al. (2023) found that some maternal mental health apps, especially those using automated psychological components, improved EPDS scores and reduced symptoms of PPD. However, most of these tools remain fragmented, lacking personalization or integration with healthcare systems, and there is limited evidence on their long-term effectiveness (Miura et al., 2023; Tsai et al., 2022).

What sets our solution apart is how it connects digital support with clinical care. Unlike general wellness apps or social platforms, our approach is grounded in maternal mental health research and built to fit within NHS services. The platform brings together clinical screening, AI-driven symptom tracking, and referral tools to support both individual recovery and broader public health efforts. By partnering with the NHS and public health teams, we aim to offer a scalable, cost-effective way to support new mothers early on. This benefits everyone: mothers receive timely support and community, healthcare providers get tools to spot issues sooner and ease their workload, families learn how to offer meaningful help, and policymakers gain a clearer, data-informed strategy to address rising mental health needs.

While barriers remain, such as data privacy concerns, the challenge of sustained user engagement, and the need for regulatory approval, our solution is uniquely positioned to overcome them by combining evidence-based design, clinical collaboration, and empathy-driven technology. By addressing not just the symptoms of postpartum depression but the root causes of isolation, we believe this innovation can make a lasting impact on maternal mental health in London and beyond.

Technical Feasibility

Existing Solutions and Gaps

Digital mental health tools are more accessible than ever, offering mood trackers, self-help apps, and online communities. But many of these tools exist in isolation and are not connected to NHS systems, which limits their ability to support ongoing care or inform clinical decisions. For example, perinatal mental health apps often do not allow meaningful data sharing with healthcare providers or provide long-term support (Eisner et al., 2022). The ClinTouch DAWN P app showed that daily mood check-ins using the Edinburgh Postnatal Depression Scale were both practical and well-received among small groups of UK mothers. Still, it remains in the pilot stage and has not been adopted into routine NHS care (Eisner et al., 2022).

Many digital mental health tools overlook new mothers' voices. Research shows that mothers and healthcare professionals want simple, flexible features like short check-ins and clear language that fit everyday life (Varma et al., 2023). Most platforms don't reflect cultural or language diversity and often lack peer support or personalized feedback to keep users engaged (Bilal et al., 2024). While some platforms offer peer support, they aren't designed for the unique emotional challenges of the postpartum period or linked to NHS care. This project aims to create an inclusive, clinically informed digital platform co-designed with mothers and NHS professionals, something women can trust, find easy to use, and feel truly connected to their healthcare.

Scientific and Operational Foundations

We know that social support, self-awareness, and timely emotional guidance are essential for maternal mental health. Many new mothers feel isolated or overwhelmed, which can lead to symptoms of postpartum depression. Encouragement and practical help from peers, family, or community members can make a real difference (Khademi, 2024). Self-monitoring

tools like mood tracking apps let women check in with themselves whenever they need to. These tools are especially helpful for those with mild to moderate symptoms or for people who may face barriers to accessing traditional mental health services, including those from underserved communities (Arean et al., 2016). They help women notice changes in their wellbeing early on and encourage them to seek support.

Artificial intelligence is also starting to play a meaningful role in mental health. Research shows that AI powered chatbots can help reduce depression and anxiety by providing conversational support and coping advice in a natural, accessible way (Fulmer et al., 2018). In this project, AI will support a friendly chatbot that guides emotional check-ins, shares evidence-based strategies, and connects users to helpful resources. It is meant to work alongside clinical care, not replace it, by encouraging early engagement and regular monitoring. With safeguards in place to detect crises and direct users to emergency support, this type of technology can help provide early intervention and more complete care for new mothers.

Resources and Activities for Development

Developing this maternal mental health platform will involve close collaboration among experts. AI developers and natural language processing specialists will build a reliable, responsive chatbot, while perinatal mental health professionals will guide the creation of clinically sound, evidence-based content. Designers will ensure the platform is accessible and user-friendly, and cultural advisors will shape an experience that resonates with women from diverse backgrounds. Clinical governance consultants will help ensure NHS data protection and ethical standards are met throughout.

To make sure the platform is accessible to as many people as possible, it will be built using React Native so it works smoothly on both iOS and Android devices. It will be securely hosted in the cloud, with all user data encrypted during transmission and storage. The platform will follow NHS Digital's Data Security and Protection Toolkit to meet privacy and healthcare data standards (Aljedaani and Babar, 2020). The team will take an Agile approach, developing the platform in phases with regular input from users. From the beginning, postpartum women and NHS stakeholders will join co-design workshops to help shape the platform and ensure it reflects real needs and lived experiences. Research shows that this kind of collaboration improves usability, cultural relevance, and equity in digital health tools (Kilfoy et al., 2024).

Once a minimum viable version is ready, pilot programs will begin with NHS trusts and maternal health organizations such as PANDAS and Maternity Action. These early trials will help assess how the platform fits into both clinical care and community settings. Local councils will be invited to support outreach and evaluation, and feedback from these pilots will guide ongoing improvements. To support development and long-term growth, the team will seek funding through NHS innovation grants, public health programs, and femtech accelerators. The project will move through clear stages, starting with understanding user needs, building prototypes, testing with mothers, and preparing for NHS integration. Table 1 outlines the timeline and key activities for each stage.

Table 1: Developmental Stages and Timeline for Maternal Health Platform

Phase	Activities	Duration
1. Research and Mapping	Conduct needs analysis, stakeholder interviews, and evidence-based research	Months 1-3
2. Prototype Development	Build core modules: chatbot + mood tracker	Months 4-7
3. User Testing and Iteration	Recruit 50-100 postpartum users in London for pilot testing	Months 8-11
4. Technical Evaluation	Monitor user retention, engagement, and symptom changes	Months 12-15
5. NHS Alignment and Scaling	Prepare documentation for DTAC compliance and NHS API integration	Months 16-18+

This table outlines how the maternal health platform develops over time, from early research to NHS integration and scaling.

Table 2 highlights the key challenges we expect during development and rollout of the maternal mental health platform. It outlines the risks involved and how we plan to address them. Our main goals are to keep users safe and engaged, honor cultural diversity, and ensure smooth integration with NHS systems. By tackling these challenges early, we aim to build a platform that truly supports mothers and stands the test of time.

Table 2. Key Implementation Challenges, Risks, and Mitigation Strategies

Risk	Mitigation
Data Security	Protect sensitive health data with strong encryption secure authentication, and NHS data standards.
AI Safety	Limit chatbot to emotional support, include emergency escalation protocols, and involve clinicians in oversight.
User Retention	Design brief, flexible activities with reminders and peer support to maintain engagement despite busy schedules.
Cultural Accessibility	Work closely with multilingual and diverse communities through co-design to ensure cultural relevance and inclusivity.

Key implementation challenges and strategies to support a safe, inclusive, and sustainable maternal mental health platform.

Market Analysis and Competitiveness

Market Context

The maternal mental health landscape in the UK, and London specifically, is experiencing increasing demand for innovative, scalable, and accessible support services. According to the NHS (2022), 10-15% of women experience PPD within the first year after giving birth. With over 106,000 live births annually in London (ONS, 2023), this suggests that approximately 13,000 to 16,000 new mothers in the capital may be affected each year. The

COVID-19 pandemic has further intensified these numbers by increasing social isolation, stress, and economic insecurity, factors strongly associated with perinatal mental health disorders (Topalidou et al., 2020).

Globally, the femtech market, defined as digital technologies targeting women's health, is projected to grow from an estimated \$50 billion in 2024 to over \$100 billion by 2030, with a compound annual growth rate (CAGR) of around 16% (Frost & Sullivan, 2024). Mental health-focused femtech, a subsegment of this market, is gaining rapid traction, with investors recognizing the unmet demand and scalability of digital interventions. This market evolution provides fertile ground for the introduction of new tools, particularly those offering holistic, culturally inclusive, and clinically sound solutions.

Problem Definition and Unmet Needs

Despite the existence of clinical screening tools and NHS perinatal mental health teams, significant service gaps remain. A report from the Maternal Mental Health Alliance (MMHA, 2022) reveals that nearly 40% of women who need mental health support during or after pregnancy do not receive it promptly. Barriers include:

- Long wait times for psychological therapy (IAPT) services.
- Cultural stigma surrounding mental illness and motherhood.
- Inadequate local provision, especially in underserved communities.
- Limited access to customized, continuous support outside of clinical settings.

These systemic gaps disproportionately affect women from low-income, minority, or immigrant backgrounds, groups overrepresented in London (Trust for London, 2023). Moreover, the UK's mental healthcare workforce is under severe strain, with psychiatrist shortages and increasing caseloads (BMA, 2024). As such, the NHS alone cannot meet the growing mental health burden without supplemental, community- and technology-based interventions.

Market Segmentation and Target Users

Our product specifically targets first-time mothers in London within six months postpartum, particularly those identified as at risk of PPD. These may include:

- Mothers aged 20–35, digitally literate and smartphone-reliant.
- Urban mothers with limited family support networks.

- Immigrant mothers experiencing cultural or language barriers.
- Mothers who are screened as “borderline” on EPDS scores in clinical settings but are not yet in acute care.

This segmentation allows for initial user profiling, testing, and scaling. Secondary user segments include partners, peer supporters, and health professionals who can use the app as an educational and monitoring adjunct. In marketing terms, this is a psychographically aligned user group motivated by community support, convenience, and confidentiality.

Competitor Analysis

The maternal mental health digital landscape in the UK is nascent, fragmented, and underserved. While several applications target aspects of new motherhood, very few offer a clinically integrated, culturally inclusive, and psychologically grounded solution.

Table 3. Structured Competitor Comparison

Competitor	Functionality	Target User Base	Pricing Model	Market Positioning	Strengths	Weakness
Peanut	Community matching, group chats, discussion forums (Peanut, n.d.)	Urban mothers, digitally savvy	Free	“Social network for mums”	High User engagement, large user base	Lacks clinical integration, no therapeutic modules
Mush	Local meet-ups, chat features (Mush, n.d.)	First-time mum, age 20-35	Free	Social matching for motherhood	Geo-based interaction, intuitive UX	No psychological support, lacks diversity features
Mum & Baby App	Maternal health education, appointment tracking (NHS NWL)	NHS patients in North West London	Free (NHS funded)	Informational tool by healthcare providers	NHS-backed, reliable information	No interactivity, no personalization or mental health focus

Headspace/ Calm	Mindfulness, meditation, stress management (Headspace, n.d.) (Calm, n.d.)	General population, stress-prone	£49.99/year+	Mainstream mental health apps	Strong brand, high-quality content	Not maternal-specific, no PPD targeting or NHS alignment
BetterHelp/ Talkspace	Online therapy with licensed counselors (BetterHelp, n.d.) (Talkspace, n.d.)	High-income, long-term therapy users	£200+/month	Digital therapy platforms	Real-time therapy, professional support	High cost, no maternal focus, limited cultural inclusivity
BabyBuddy	Health education videos, milestone reminders (BabyBuddy, n.d.)	Low-income or ethnically diverse mothers	Free (Charity-funded)	Public health educational tool	Multilingual, equity-oriented	No interactive or personalized mental health features

Table 4. Market Share (UK-based Estimate)

Platform	Est. Monthly Active Users (UK)	Estimated Market Share in Maternal Digital Health
Peanut	~300,000	~35% (Peanut, n.d.)
Mush	~150,000	~18% (Mush, n.d.)
BabyBuddy	~80,000	~10% (BabyBuddy, n.d.)
Headspace/ Calm	~250,000 (non-specific)	N/A (non maternal-specific) (Headspace, n.d.; Calm, n.d.)
Our APP (Project Year 3)	10,000-20,000	~5% (in target regions such as Greater London)

Note: Market share figures are derived from publicly available user estimates, third-party reports, and projected adoption numbers from our financial model. Exact figures may vary due to proprietary data limitations.

Insight: While apps like Peanut and Mush dominate user engagement in social features, they lack clinical depth. No app currently claims dominant share in the maternal mental health intervention niche, especially one built for NHS integration and multicultural engagement.

Value Proposition and Unique Selling Points (USPs)

Our digital platform offers multiple differentiators that directly respond to market needs:

- Clinical + Community Model: Combining evidence-based CBT modules, AI-driven emotional support, and peer networking in a single app.
- NHS Integration Readiness: Aligned with DTAC and NICE evidence frameworks for public adoption.
- Cultural Inclusivity: Developed with input from BAME and migrant mothers to include multilingual content, flexible access, and culturally relevant modules.
- AI Chatbot and Crisis Escalation: Real-time AI mood tracking and crisis escalation protocols.
- Holistic Wellness Modules: Includes recipes, guided fitness classes, breathing exercises, and infant-care tips to promote physical and mental recovery.

Channels and Partnerships

Key partnerships will drive user acquisition, retention, and credibility:

- Hospitals and NHS Trusts: For early identification, referrals, and subsidized access. Co-branded rollouts enhance institutional trust.
- Charities and NGOs: Including MumsAid, Mothers Matter, PANDAS, and the Maternal Mental Health Alliance—critical for outreach, co-design, and user testing.
- GP Practices and Health Visitors: Can recommend the app during the 6-week postnatal check-up.
- Local Authorities: For grant funding and integration into maternal well-being programs.
- Femtech Accelerators: Such as FemTech Lab or DigitalHealth.London for mentorship, funding, and investor visibility.

Business Viability

Regulatory Considerations

Our digital platform for maternal mental health must follow strict rules around data protection and health technology if it is to be used in the UK healthcare system, particularly within the NHS.

Data Protection and Privacy

Because the app collects sensitive health data such as mood tracking and chat records, it must fully comply with the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018 (ICO, 2023). Key principles underpinning our approach include ensuring informed consent, whereby users provide clear and explicit permission for data usage. We will strictly adhere to data minimisation, collecting only the information essential to service delivery. All data will be encrypted during both storage and transmission to maintain high security standards. Furthermore, user privacy will be prioritised through the use of anonymised or pseudonymised data wherever possible. To ensure sustained compliance, a designated Data Protection Officer (DPO) will oversee all privacy-related practices.

NHS Digital Compliance (DTAC)

To be used within NHS infrastructure, our platform must meet the Digital Technology Assessment Criteria (DTAC) (NHS England, 2021). This includes fulfilling safety risk assessments in accordance with DCB0129 and DCB0160 standards, demonstrating strong data protection and cybersecurity protocols, and ensuring accessible design aligned with WCAG 2.1 AA standards. In addition, the platform must exhibit technical readiness to integrate with existing NHS digital systems. We will collaborate with NHS Innovation Hubs early in development to ensure these criteria are met.

Medical Device Regulations

Should future iterations of the platform incorporate clinical assessments, such as PPD risk scoring, it may fall under the scope of a medical device as defined by the UK Medical Device Regulations 2002 (MHRA, 2002). In that case, the app would require formal approval and UKCA marking (GOV.UK, 2021). However, the current version is designed to support, not replace, clinical care and is not classified as a regulated medical device.

Legal Framework

Relevant Laws

Our platform operates within the boundaries of several key UK laws. The Health and Social Care Act 2012 underpins services that aim to improve patient outcomes, aligning with our objective of enhancing maternal mental health (UK Government, 2012). The Children and Families Act 2014 emphasises maternal well-being, directly supporting our user group (UK Government, 2014). The Equality Act 2010 ensures non-discriminatory access, which is particularly relevant for serving minority and migrant mothers (UK Government, 2010). Additionally, the evolving Online Safety Act may impose future responsibilities regarding AI moderation and online content governance (Ofcom, 2023).

Health Policy Alignment

Our solution is also well aligned with national healthcare strategies. The NHS Long Term Plan (2019) advocates for increased use of digital tools in perinatal mental health care (NHS, 2019). The Women's Health Strategy for England (2022) places maternal mental health as a core priority (Department of Health and Social Care, 2022). Similarly, London's Digital Health Strategy (2023) promotes the adoption of technology-enabled services targeted at vulnerable populations (Greater London Authority, 2023). This strategic alignment enhances our likelihood of receiving institutional support or public sector funding.

Legal Pathways for Growth

We intend to pursue several legal and policy-based routes for scaling. These include applying for NHS innovation funding streams such as SBRI Healthcare, NIHR, and Innovate UK (SBRI Healthcare, 2023; NIHR, 2023; Innovate UK, 2023). We will also explore opportunities through public procurement mechanisms as outlined in the UK Public Contracts Regulations 2015 (UK Government, 2015). Local collaborations with councils and third-sector organisations will further support delivery and scaling.

Intellectual Property (IP)

Our IP Assets

The protection of intellectual property is essential to maintaining our platform's competitive edge. We plan to register our brand identity, including the app name and logo, as trademarks with the UK Intellectual Property Office. All creative content, such as visual design, written materials, and therapy frameworks, will be safeguarded through copyright. Our proprietary AI algorithms, including crisis detection models, will be treated as trade secrets and

protected through robust internal governance and non-disclosure agreements. If we develop novel technologies in emotional tracking or behavioural analytics, we will assess their eligibility for patent protection under UK law.

Using IP for Business Advantage

Our IP strategy is also a vehicle for growth. We intend to offer exclusive licenses to public sector bodies, such as NHS Trusts or local councils, for co-branded implementations. In research collaborations, we will set clear terms to retain ownership of core IP while encouraging innovation. For broader impact, we may share non-core elements as open-source tools, while keeping key innovations protected to ensure a sustainable competitive advantage.

Managing IP Risks

To minimise legal and reputational risks, we will conduct thorough IP audits to avoid infringing on third-party rights. All external content and open-source tools will be carefully vetted. Legal agreements will be employed to secure intellectual property developed in partnership settings, ensuring clear ownership and use rights.

Financial model

Financing Structure

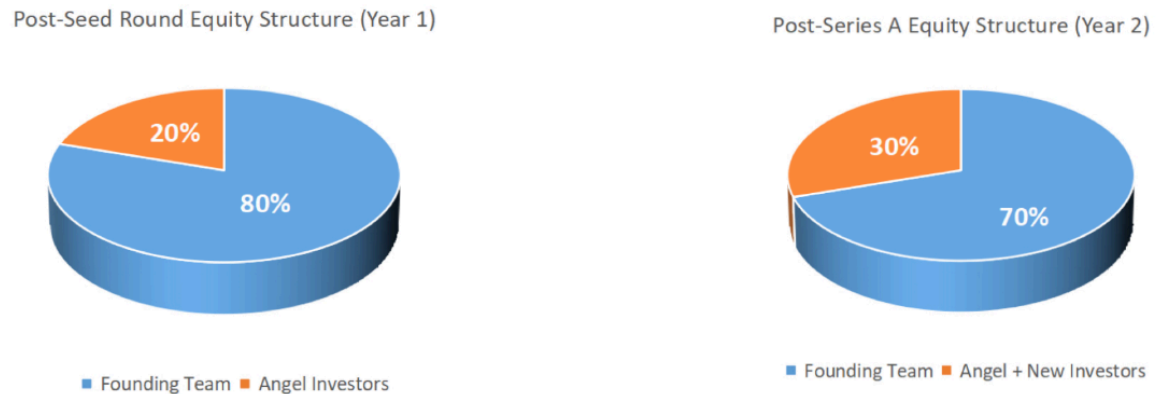
In Year 1, the company successfully raised a total of £400,000 in seed round funding to support the early-stage product development and pilot testing of the project, as well as the building of the core team. This round of financing included a £150,000 grant funding from the NHS Innovation Fund, a £200,000 investment from angel investors in exchange for 20% of the company's equity, and £50,000 in support from the Digital Health Accelerator. After the first round of financing, the founding team's ownership was diluted to 80%.

In Year 2, to support the optimization and market promotion of the product, the company conducted a second round of financing, selling 10% of its equity and obtaining £250,000 from a combination of existing and new investors.

As of now, the company has raised a total of £650,000 in financing, including £450,000 in equity investment obtained by selling 30% of its equity, as well as

government and industry support grants for the project. The founding team holds 70% of the equity (Figure 1). These two rounds of financing have also provided the necessary financial support for the early-stage research and development and market promotion of the product.

Figure 1: Equity Structure



Revenue Forecast

The main revenue models of this business project consist of two key components:

The first is a subscription-based model for individual users. On the basis of enjoying free functions, users can pay £50 to subscribe to the advanced cognitive behavioral therapy module or personalized coaching services.

The second service is a licensing model for NHS partner institutions. Each NHS institution pays an annual authorization fee of £10,000 and can apply the platform services in the medical process of patients to provide mental health support for perinatal women. The projected revenue over the next five years is illustrated in Appendix 1.

User Growth and Retention Forecast

For the projected growth of user numbers within the first five years after the project's launch, considering that the main target group is perinatal depressed women, there is a risk of user attrition as they recover postpartum and their mental state improves.

Based on industry benchmarks and the advantages of our product, we estimate that the annual user retention rate can reach up to 60%. Research shows that for mental health apps without human-assisted functions, the user retention rate within 30 days is only around 4% (Kaveladze et al., 2022). However, for similar products with human feedback or counseling functions, the retention rate can reach about 50% (Torous et al., 2020). Our platform not only features AI dialogue, community gatherings, 24/7 medical consultation services and other functions, but also collaborates with the NHS system to enhance users trust and long-term engagement. Meanwhile, many users continue to use the service from 6 months to 1.5 years after giving birth. Therefore, the final annual retention rate is set at 60%. Due to the high trust and strong communication networks among new mothers, we assume a 10% annual referral conversion rate. Positive experiences and improvements in mental health motivate many users to share the product with other pregnant or postpartum women in their communities.

Cost Forecast

The main costs of this project include fixed expenses, such as employee salaries and office rent, and variable costs such as marketing expenses and cloud service fees (Appendix 2 and Appendix 3). As shown in Appendix 4, the core team consists of five members in the first three years, covering key positions such as product development, clinical advisors, and operations. As revenue grows, the company plans to hire three more employees in Year 4, including a product manager, a data analyst, and a public sector partnership manager, to support business expansion (Office for National Statistics, 2024).

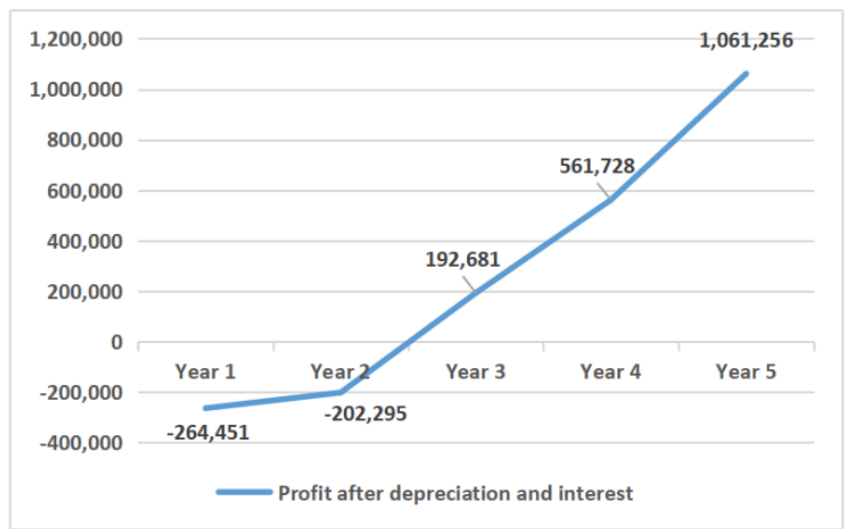
To control costs, the company adopts a flexible office model, choosing shared office spaces in London with rent costs controlled at £300 to £500 per person per month (WeWork, 2025). At the same time, it supports remote and online working models to reduce fixed expenses.

Income and Profitability Projection

Over the five-year forecast period, the total revenue of this business project increases from £40,000 in Year 1 to £1,700,000 in Year 5, with a compound annual growth rate of approximately 156.3%, as shown in Appendix 5. As shown in Figure 2, EBITDA turned positive in the third year, and the net profit changed from a loss of £264,451 in Year 1 to a profit of £1,061,256 in Year 5. This indicates that as the market penetration rate increases and service usage continues to grow, the project gradually turns losses into profits, driven by scale effects

and improvements in operational efficiency. These results demonstrate strong profitability potential over time.

Figure 2 : 5-Year Profit After Depreciation and Interest



Financial Forecast Overview

The projected financial performance of the company is reflected through five-year cash flow and balance sheet forecasts. As shown in Appendix 6, the company maintains a good cash position throughout all years, with the cumulative ending balance increasing significantly from £176,469 in the first year to £1,811,879 in the fifth year, indicating the company's effectiveness in revenue generation and cost management. Meanwhile, Appendix 7 highlights the continuous strengthening of the company's financial structure, with total assets nearly doubling and shareholders' equity turning positive in the fourth year. These trends suggest the company's long-term financial sustainability and readiness for future growth.

Strategy to Market and Implementation

Market Strategy

Based on the previous sections, our product's core mission is to provide mothers experiencing postpartum depression with accessible, clinically evidenced mental health support anytime and anywhere, through a dual-track intervention model combining human community and AI communication. Key strategic differentiators are 24/7 accessibility and clinical evidence. As discussed in the competitive analysis, most products offer only online communities and lack robust clinical evidence. We address the issue of anytime accessibility through our AI chatbot and plan to conduct real-world research to provide future validation.

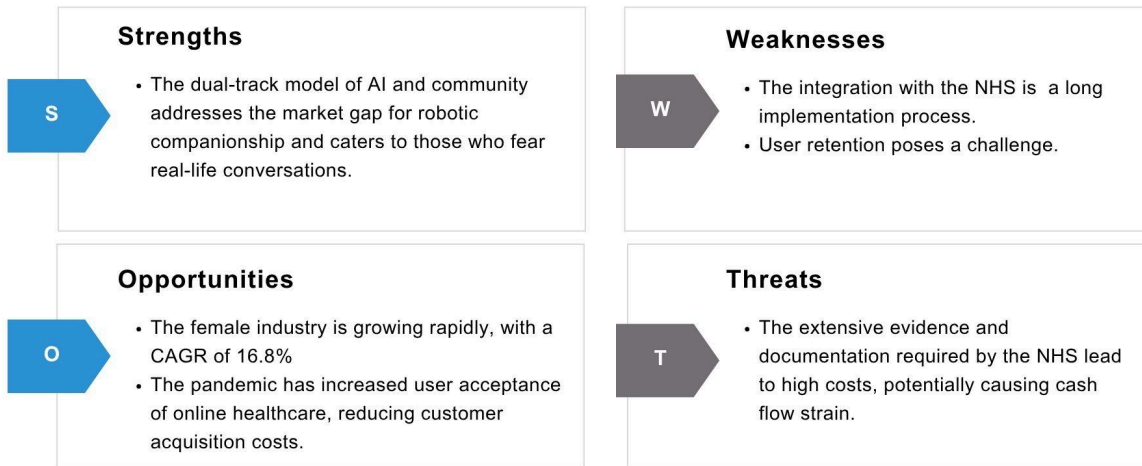
For market entry, we primarily pursue a business-to-business-to-customer model by partnering with the NHS. To avoid a single channel, we also use a direct-to-consumer model to collaborate with offline community events to expand the user base. Our NHS partnership strategy prioritizes clinical validation to demonstrate clinical and economic effectiveness, which will support large-scale collaborations and further expansion. The following section presents a SWOT analysis, briefly discusses the weakness and threat, and shows solutions in the subsequent milestones.

SWOT Analysis

Regarding market positioning, our advantage is the dual-track model of AI and community. The AI component fills a gap for mothers avoiding real human interaction. Studies show perinatal depression leads women to isolate themselves due to lack of support (Adlington et al., 2023). And our AI robot can help them release emotions and provide companionship. However, weaknesses exist. NHS integration involves complex procedures and lengthy approvals, delaying commercialization and increasing initial costs. User retention is another challenge. Target users often have limited time due to caregiving responsibilities, reducing app engagement. Even after symptom relief, users may drop off. Although previous sections mention that our features can extend usage, this still requires attention.

Still, the growing female health market presents an opportunity, projected to reach \$3785.3 million in 2030 with a 16.8% CAGR (HORIZON, n.d.). The pandemic also increased users' acceptance of digital health. Nevertheless, NHS regulations remain a threat. Its requirements for trials and certifications increase the cost budget and may cause funding shortage and failure. SWOT is shown in Figure 3.

Figure 3 SWOT Analysis



Despite the challenges of a complex NHS access process, a long initial pilot period, and difficulty in user retention, there are clear paths to address these issues. In response to the slow pace of NHS collaboration and high evidence requirements, we developed the product in strict accordance with the DTAC standards and prepared relevant technical and evidence materials in advance to speed up the approval process. Additionally, our product belongs to the Tier B category in the NICE digital health standards, and the evidence requirements that need to be met can be gradually completed through real-world studies(RWS) and budget impact analysis(NICE, 2018). To address the issues of retention users, it is necessary to analyse product data and formulate corresponding retention product functions or strategies. Based on this, we have developed a phased implementation roadmap to ensure the steady progress of the product at the regulatory, clinical and user levels.

Roadmap and Milestones

The project roadmap includes three stages. The first stage is the minimal viable product (MVP) development and pilot stage, from June 2025 to June 2026. Development will strictly obey NHS standards, adopting MVP strategy with different batch feature releases. Core features are built first, while other community functions are added iteratively. A pilot will be conducted in one London area in partnership with a local GP. Concurrently, DTAC preparations will begin, including gathering developing materials and evidence. The key tasks for this phase

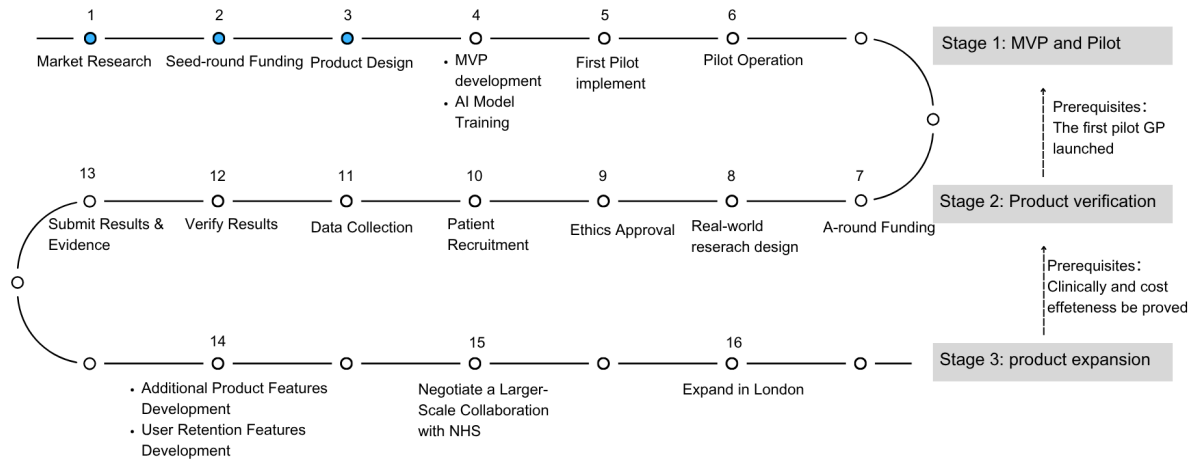
include completing core functions, AI model training and internal testing by April 2026, as well as establishing GP partnership and deployment training. These milestones are essential to evaluate the phase's success and move forward. While our primary goal is to conduct a clinical pilot with one GP site, we will simultaneously engage in commercial discussions with additional GP practices to ensure early-stage cash flow. Accordingly, our Year 1 target is to secure contracts with two GP partners.

The second stage, from June 2026 to March 2027, is the product verification stage. A small-scale RWS will be conducted to verify the AI function's accuracy, its impact on postpartum depression, and the economic benefits. Evidence will be prepared and submitted to the NHS to facilitate further expansion. The RWS will last nine months, user engagement data will be collected, recovery progress from patient follow-ups will be analysed, and GP workload for the trial participants will be measured. These metrics will then be compared against the UK's historical averages for postpartum depression recovery time and average GP workload to determine statistical significance. Finally, evidence documents will be submitted in accordance with the standards of NICE Tier B. This phase's key indicator is whether the trial results have economic or clinical benefits.

The third stage is the product expansion stage, which will be after March 2027. This stage involves formal collaboration with the NHS and expansion across all of London. Meanwhile, all remaining product features will be developed. During this period, user retention data will be monitored to dynamically adjust the development priority of features. If the retention data is poor, we will develop the retention-related functions first. Otherwise, the development may be postponed. Important metrics for measuring this stage are that the monthly active users (MAU) must be greater than 1,0000 and the six-month user retention rate must be greater than 60%.

It must be acknowledged that even if we develop and prepare evidence strictly in accordance with NHS regulations, collaboration with the NHS may not be as smooth as we imagine. Therefore, our contingency plan is to acquire users through collaboration with individual doctors. As mentioned above, we also have customer acquisition methods through community collaboration. At the same time, we will add direct-to-consumer customer acquisition and expand our user base through targeted advertising to women. Within this approach, we will continue to conduct RWS trials to collect evidence, which will not only serve as advertising material for attracting users, but also lay the groundwork for future collaboration with the NHS. Detailed milestones are shown in Figure 4.

Figure 4: Detailed Milestones



Collaboration ecosystem and risk Hedging

Risk hedging in a digital health startup is the thoughtful manipulation of uncertainty that may pose a hazard to the business's financial, operational, or regulatory security. Because of various challenges involved in deploying a maternal mental health app in the National Health Service (NHS) in the UK, mitigation of risks should be proactive to guarantee future sustainability (Reidy et al., 2025). Creating a strong collaboration ecosystem that combines resources, skills, and networks to overcome hurdles and increase impact (Aksoy, 2023). The well-organised ecosystem reduces risks and enables faster growth by relying on unions with governmental organisations, charities, and technology companies.

Among the significant risk strategies used in this project are revenue diversification. The app eliminates the risk of relying on revenue from a single source by having a mixed business model (institutional licensing, such as NHS contracts, freemium subscriptions, and corporate partnerships). This will create financial risks, especially when the procurement process in the public sector may be slow. Regulatory risks may be eliminated with the cooperation of the legal and consulting companies specialising in NHS compliance. These specialists can facilitate the

authorisation procedure and prevent the blockage, which means the timely implementation. The other important strategy is the gradual implementation of the app, i.e., pilot testing in the boroughs of high needs in London and the rest of the country. This concurs with the Minimal Viable Product (MVP) approach; the subsequent iterative enhancements may be made per the feedback obtained in the real world, but the aspects of financial exposure on a large scale are reduced.

Involving the NHS Innovation Hubs can be mentioned as one of the steps to take early in the development process because it aligns with the healthcare system's priorities and eases the integration process with the already established perinatal care pathways (NHS England, 2019). Moreover, joint charity work (PANDAS and MumsAid) will support grassroots engagement, overcoming cultural stigma and increasing the uptake of the vulnerable groups (Maternal Mental Health Alliance, 2022). On the technology side, alliances with cloud service providers, such as Amazon Web Services (AWS), guarantee scalable infrastructure.

Forecasted Return on Investment

The projected financial viability of this maternal mental health application is evaluated using a sound analysis of finances. This forecast includes license-based NHS and premium subscription revenue. It assumes a compound growth rate (CAGR) of 16% used by the femtech industry to indicate longer projections (Avalere Health, 2024). Such an approach would allow for consistency with the market trends, considering the peculiarities of the digital health intervention in the UK setting.

The primary monetary measures show high investment opportunities for the project. Having only a capital requirement of 500,000, representing the initial development cost and regulatory compliance cost, the venture is also estimated to break even after the first three years of operation. Using the net present value (NPV) measurement with a conservative discount rate of 8%, it is estimated that significant financial sustainability is within a five-year planning period, as it forecasts an approximation of 1.2 million in financial feasibility. Moreover, the expected development of 3.5 years is calculated with the constant extension of the number of users by 10% per annum, which means realistic growth levels in the market penetration of the target population in London (Statista, 2024).

A detailed sensitivity analysis will consider the fluctuating conditions of the market. Based on the optimistic scenarios where 15% of the postpartum population in London is

estimated to use the service by Year 3, revenues of up to pounds 750,000. In a more pessimistic scenario with slower rates of 5% adaptations and delays in the procurement process of the NHS, annual revenues would drop to 200,000, and the break-even would be reached in the fifth year. These forecasts emphasise the need for strategic NHS affiliation and efficient user acquisition.

The economic returns are not the only value offered; the application could generate significant social impact. The NHS is estimated to spend 1.2 billion annually addressing postpartum depression (PPD) (UK Government, 2019). Once adopted in the UK, the solution has potential to scale into EU markets due to harmonised regulations and similar healthcare systems.

Strategic risks and risk mitigation measures

Effective deployment of the maternal mental health application involves several strategic risks that must be addressed appropriately. One key issue relates to regulatory compliance, particularly adherence to the NHS Digital Technology Assessment Criteria (DTAC), which can require over 18 months for approval (NHS England, 2021). Another concern is user engagement, as new mothers may show low retention rates, a common issue with digital health tools (Baumel et al., 2019). Data security is also critical, with potential GDPR violations carrying penalties of up to 4% of annual revenue (UK GDPR, 2021), particularly concerning given the sensitivity of mental health information.

A holistic mitigation strategy has been proposed. A practical regulatory pathway will be supported by experienced NHS compliance consultants familiar with navigating DTAC, based on successful past implementations under the NHS Long Term Plan. Evidence-based engagement strategies—such as gamification features, achievement rewards, and personalised push notifications—will help improve retention, as shown in other mental health applications (Miura et al., 2023). Data protection will be strengthened through ISO 27001 certification and annual third-party audits, ensuring alignment with the Data Protection Act 2018.

Long-term sustainability also depends on adapting to shifts in the healthcare environment. Changes in NHS policy, including potential privatisation trends, will be monitored by reviewing British Medical Association reports to develop proactive responses (BMA, 2024).

These steps contribute to a dynamic risk management approach that balances near-term operations with long-term strategic flexibility. The application will be well-positioned to adapt and grow within the rapidly evolving digital health market.

Future Work

Future work will prioritize large-scale clinical validation to assess the app's long-term impact on maternal mental health across diverse populations. We aim to optimize AI responsiveness and integrate personalized care pathways based on user profiles. The project will also expand geographically across the UK, introduce multilingual support, and explore cross-border deployment. Partnerships with public health agencies, academic institutions, and digital health accelerators will play a key role in future scaling. Additionally, we will continue refining data privacy protocols and ethical safeguards to ensure compliance with evolving regulatory standards and user trust.

To achieve the above goals, we will carry out the following activities. Regarding clinical validation, after demonstrating the clinical and economic efficacy of the product in the first trial of the second year, we plan to focus on advancing collaborations with the NHS to conduct larger-scale trials, including expanding the population size, ethnic diversity, and regions covered. This will provide foundational data for the future scalability of the product. In terms of product functionality, improving the AI robot's response accuracy, emotion recognition capabilities, and interaction personalization will remain our top priorities. We will dispatch product managers to conduct follow-up visits with users and dynamically adjust the AI model parameters based on real-world feedback and actual usage data. Furthermore, once the general AI model is stable, we will develop personalized AI models. These personalized models will recommend differentiated emotion guidance content to users with different personalities, religious beliefs, and ethnicities, enabling more targeted interventions. For market expansion, we will launch multilingual versions to enhance market feasibility such as Chinese, Arabic, and Spanish. In the future, we will further expand our partnerships, such as collaborating with academic institutions to further develop an upgraded version of our CBT curriculum. Additionally, we will collaborate with more partners like gyms and book clubs. These partners will also help us expand our network and thus expand our market. Finally, ethics and data privacy mechanisms are always our top priorities. As the market expands, we will continuously

upgrade them based on local ethics and data privacy mechanisms. At the same time, we will conduct both machine and human review of sensitive terms and violations in conversations on the platform. If sensitive terms appear, users will be notified, thereby enhancing the platform's social credibility and professionalism.

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Appendix

Appendix 1: 5-Year Revenue Forecast

	Year Annual Profit and Loss Account					
<u>Revenue</u>	Year 1	Year 2	Year 3	Year 4	Year 5	TOTAL
Service 1						
Number of clients	400	2,000	10,000	20,000	30,000	62,400
Service price	£50	£50	£50	£50	£50	£250
Subtotal revenue	£20,000	£100,000	£500,000	£1,000,000	£1,500,000	£3,120,000
Service 2						
Number of NHS clients	2	5	10	15	20	52
Service price	£10,000	£10,000	£10,000	£10,000	£10,000	£50,000
Subtotal revenue	£20,000	£50,000	£100,000	£150,000	£200,000	£520,000
Total Revenue	£40,000	£150,000	£600,000	£1,150,000	£1,700,000	£3,640,000

Appendix 2: 5-Year Marketing Expenses Forecast

	5 Year Annual Profit and Loss Account					
<u>Marketing expenses</u>	Year 1	Year 2	Year 3	Year 4	Year 5	TOTAL
Digital Advertising	£6,000	£12,000	£18,000	£22,000	£25,000	
Social Media	£4,000	£6,000	£10,000	£12,000	£15,000	
Branding & PR	£5,000	£5,000	£12,000	£6,000	£6,000	
Outreach & Events	£2,000	£4,000	£6,000	£6,000	£7,000	
Marketing Research	£2,000	£3,000	£4,000	£4,000	£5,000	
Total Marketing Expenses	£19,000	£30,000	£50,000	£50,000	£58,000	£207,000

Appendix 3: 5-Year Cost of Sales Forecast

	5 Year Annual Profit and Loss Account					
<u>Cost of Sales</u>	Year 1	Year 2	Year 3	Year 4	Year 5	TOTAL
	£	£	£	£	£	
Cloud & AI	3500	6000	10000	12500	15000	£47,000
Clinical review	4000	6000	8000	10000	12000	£40,000
Total cost of sales	£7,500	£12,000	£18,000	£22,500	£27,000	£87,000

Appendix 4: 5-Year Staff Costs Forecast

5 Year Annual Profit and Loss Account											
	Start of employment (Y n, M n)	Year 1		Year 2		Year 3		Year 4		Year 5	
		Salary Gross	NI er	Salary Gross	NI er	Salary Gross	NI er	Salary Gross	NI er	Salary Gross	NI er
		£	£	£	£	£	£	£	£	£	£
Tech Lead (AI & Backed)	Y1 M1	56,000	6,365	60,000	6,917	62,000	7,193	66,000	7,745	70,000	8,297
Clinical Consultant (Perinatal Mental Health)	Y1 M1	42,000	4,433	45,000	4,847	49,000	5,399	52,000	5,813	55,000	6,227
Front-end Developer (React Native)	Y1 M4	40,500	4,226	46,000	4,985	51,000	5,675	54,000	6,089	58,000	6,641
UX/UI Designer	Y1 M4	36,000	3,605	41,000	4,295	45,000	4,847	49,000	5,399	53,000	5,951
Marketing Staff	Y1 M4	24,000	1,949	27,000	2,363	30,000	2,777	33,000	3,191	36,000	3,605
Product Manager	Y4 M1	-	-	-	-	-	-	50,000	5,537	53,000	5,951
Data Analyst	Y4 M1	-	-	-	-	-	-	39,000	4,019	44,000	4,709
Public Sector Partnership Manager	Y4 M1	-	-	-	-	-	-	42,000	4,433	45,000	4,847
Total gross pay		198,500	20,576	219,000	23,405	237,000	25,889	385,000	42,222	414,000	46,224
NI employer allowance		-	5,000	-	5,000	-	5,000	-	5,000	-	5,000
Total annual (inc NI employer)		214,076		237,405		257,889		422,222		455,224	
Pension contributions employer		5,955		6,570		7,110		11,550		12,420	
Total employment costs		220,031		243,975		264,999		433,772		467,644	

Appendix 5: 5-Year Profit and Loss Account

Year Annual Profit and Loss Account						
	Year 1	Year 2	Year 3	Year 4	Year 5	TOTAL
Revenue	£40,000	£150,000	£600,000	£1,150,000	£1,700,000	£3,640,000
Cost of Sales	£7,500	£12,000	£18,000	£22,500	£27,000	£87,000
Gross Profit	£32,500	£138,000	£582,000	£1,127,500	£1,673,000	£3,553,000
<u>Overhead Expenditure</u>						
Operating expenses						
Payroll costs	£220,031	£243,975	£264,999	£433,772	£467,644	£1,630,421
Serviced office rent	£36,000	£36,000	£36,000	£38,400	£38,400	£184,800
Software subscription	£900	£1,200	£1,400	£2,200	£2,200	£7,900
Legal expense	£3,000	£3,000	£5,000	£5,000	£5,000	£21,000
Marketing expense	£19,000	£30,000	£50,000	£50,000	£58,000	£207,000
Office Expenses	£1,500	£2,000	£2,000	£2,500	£3,000	£11,000
Accountancy Fees	£2,000	£2,000	£2,000	2,000	2,000	£10,000
Insurance	£2,000	£2,000	£2,000	2,000	2,000	£10,000
Business support	£1,000	£2,000	£3,000	2,500	2,500	£11,000
Miscellaneous	£654	£782	£1,014	£1,046	£1,131	£4,627
Total Overhead Expenditure	£285,431	£322,175	£366,399	£538,372	£580,744	£2,093,121.36
Total Expenditure	£292,931	£334,175	£384,399	£560,872	£607,744	£2,180,121
EBITDA	£(252,931)	£(184,175)	£215,601	£589,128	£1,092,256	£1,459,879
EBITDA Margin %	-632%	-123%	36%	51%	64%	
Net Present Value @ discount rate 3.5%	-£244,378	-£177,947	£208,310	£569,205	£1,055,319	
Depreciation & Amortization	£11,520	£18,120	£22,920	£27,400	£31,000	£110,960
Profit after depreciation and interest	-£264,451	-£202,295	£192,681	£561,728	£1,061,256	£1,348,919

Appendix 6: 5-Year Cash Flow Forecast

		5 Year Cash Flow Forecast				
		Year 1	Year 2	Year 3	Year 4	Year 5
		£	£	£	£	£
Opening Balance as at		0	176,469	189,294	330,896	827,623
CEO investment*		100,000	0	0	0	0
Equity/Cash Introduced (BS)		400,000	250,000	0	0	0
Sales		30,000	130,000	550,000	1,080,000	1,610,000
Total Receipts		530,000	380,000	550,000	1,080,000	1,610,000
Cost of sales		7,500	12,000	18,000	22,500	27,000
Total Overhead Expenditure for Cash flow						
Forecast from P&L		285,431	322,175	366,399	538,372	580,744
1 month deposit for rent		3,000	0	0	0	0
Fixed Assets and R&D investment		57,600	33,000	24,000	22,400	18,000
Total expenditure		353,531	367,175	408,399	583,272	625,744
Cash Surplus / Deficit		176,469	12,825	141,601	496,728	984,256
Closing Balance		176,469	189,294	330,896	827,623	1,811,879

Appendix 7: 5-Year Balance Sheet

Balance Sheet	Year 1	Year 2	Year 3	Year 4	Year 5
	£	£	£	£	£
NON-CURRENT ASSETS					
Computers & Equipment	10,080	7,560	5,040	5,240	2,040
Intangible Assets	36,000	53,400	57,000	51,800	42,000
Non-current Assets	46,080	60,960	62,040	57,040	44,040
CURRENT ASSETS					
Cash at Bank	176,469	189,294	330,896	827,623	1,811,879
Trade receivables	10,000	20,000	50,000	70,000	90,000
Current Assets	186,469	209,294	380,896	897,623	1,901,879
TOTAL ASSETS	232,549	270,254	442,936	954,663	1,945,919
Long term Liabilities					
Director's loan account	100,000	100,000	100,000	100,000	100,000
Other Long-Term Liabilities	31,752	50,952	49,352	45,352	39,752
Total Long term liabilities	131,752	150,952	149,352	145,352	139,752
Current Liabilities					
Accounts Payable	166,698	267,498	259,098	238,098	208,698
Accured Expenses	182,574	292,974	283,774	260,774	228,574
Other Current Liabilities	15,876	25,476	24,676	22,676	19,876
Total Current Liabilities	365,148	585,948	567,548	521,548	457,148
TOTAL LIABILITIES	496,900	736,900	716,900	666,900	596,900
CAPITAL					
Called Up Share Capital	100	100	100	100	100
Share Premium	200,000	450,000	450,000	450,000	450,000
Profit and Loss Account	-264,451	-466,746	-274,064	287,663	1,348,919
TOTAL EQUITY	-264,351	-466,646	-273,964	287,763	1,349,019
TOTAL LIABILITIES & EQUITY	232,549	270,254	442,936	954,663	1,945,919
Balance Check	0	0	0	0	0